Applied Analytics / Business Intelligence Case Study

# Business Diagnostics Meets Applied Analytics:
## Turning Operational Data into Strategic Intelligence

*A case study on multidimensional organizational maturity analysis*

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**Business Intelligence & Data Analytics Case Study**
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**#DataAnalytics #BusinessIntelligence #FinancialDataAnalysis #AppliedAnalytics #Python #PowerBI #BusinessConsulting**

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## ■ Project Context

This anonymized case study shows how business diagnostics can be translated into structured analytical intelligence. An organizational maturity framework was applied across **5 business dimensions** — Strategy, Finance, Marketing, Operations, and People Management — transforming field data into actionable insights. Developed as an Applied Analytics / Business Intelligence project aligned with Big Data Analytics studies.

## ■■ Data Sources & Structure

Data collected via structured interviews, document analysis, and direct observation. Each dimension scored from **0–100%**, producing a multivariate dataset with **9 core analytical variables** across business dimensions and regional units. Anonymized financial records were also analyzed — cost distribution, fixed costs, and liquidity indicators — generating a **feature-ready dataset prepared for future predictive analytics expansion**. A synthetic version was created for reproducibility, with no real client data included.

## ■ Technical Execution

**Tools**
Python | Pandas | NumPy | Power BI

**Processing**
Data Cleaning | Normalization | Encoding

**Analysis**
Clustering | Descriptive Stats | Gap Analysis

Raw data was ingested, cleaned, and normalized using **Python (Pandas, NumPy)**. Categorical indicators were encoded for analytical compatibility. Descriptive statistics and gap analysis were computed per dimension. K-Means clustering was applied to identify maturity profiles — optimal **k determined via Elbow Method and silhouette score (0.61)** — and results were structured into a **scalable analytical schema suitable for future integration with larger datasets and BI dashboards**.

**Code — clustering_analysis.ipynb**

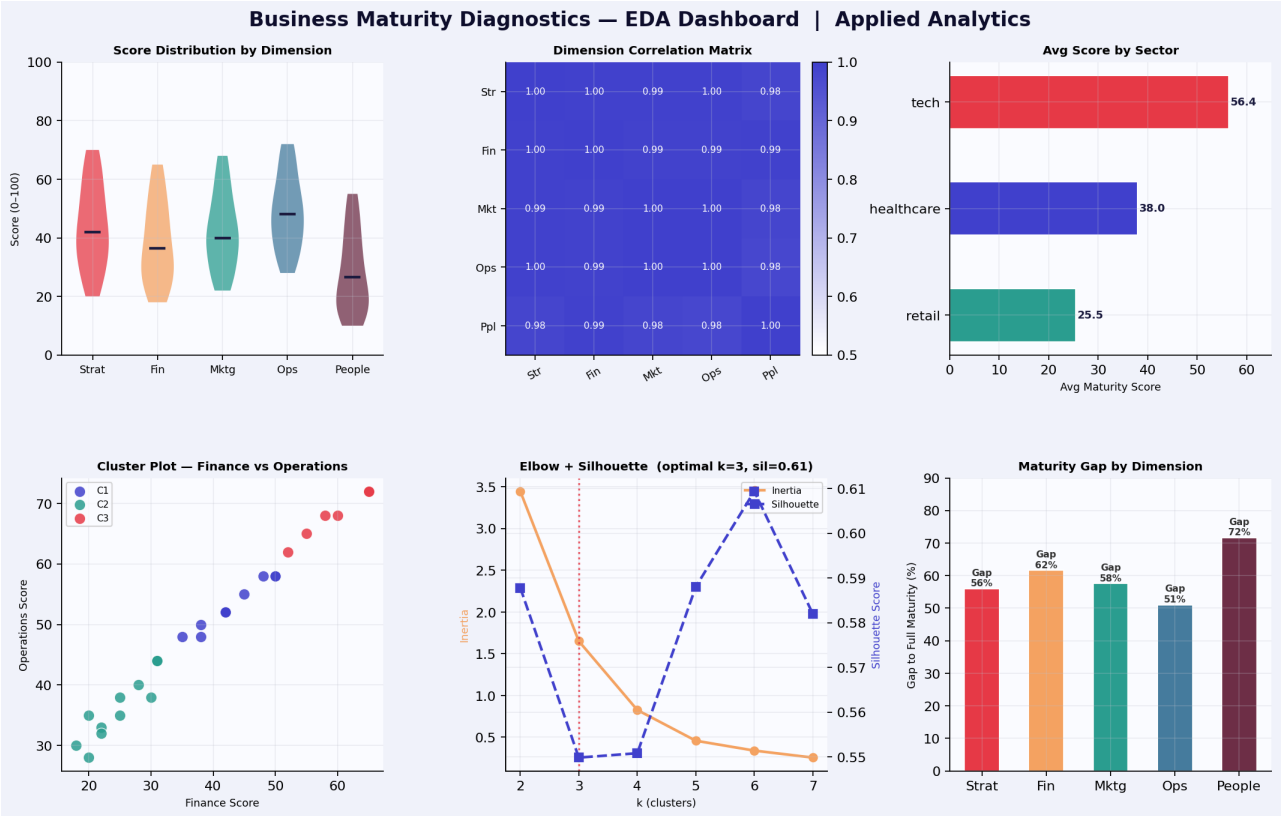
```python
clustering_analysis.ipynb                                   Python 3.11 | sklearn 1.4
import pandas as pd;  import numpy as np
from sklearn.cluster import KMeans;  from sklearn.metrics import silhouette_score

# Load and normalize maturity score matrix
df = pd.read_csv('maturity_scores_anon.csv')
X = (df[features] - df[features].min()) / (df[features].max() - df[features].min())

# Elbow method + silhouette to select optimal k
inertia, sil = [], []
for k in range(2, 7):
    km = KMeans(n_clusters=k, random_state=42)
    labels = km.fit_predict(X)
    inertia.append(km.inertia_)
    sil.append(silhouette_score(X, labels))

# Fit final model with optimal k=3
km_final = KMeans(n_clusters=3, random_state=42).fit(X)
df['cluster'] = km_final.labels_
print(f'Silhouette: {silhouette_score(X, df["cluster"]):.3f}')
df.groupby('cluster')[features].mean().round(1)
```

### EDA Dashboard — Python / matplotlib



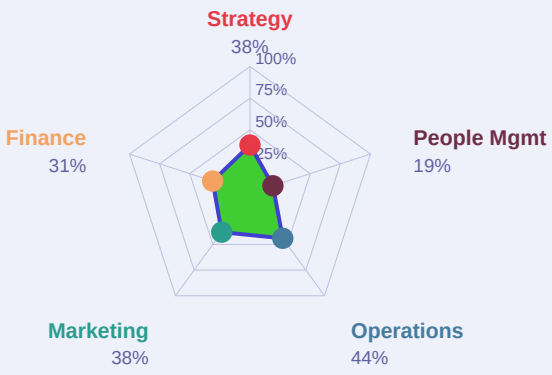

### Repository

**Public repository: synthetic dataset, clean notebook, and README — no real data.**

*Post link only when repo is public, clean, and verified: README · synthetic dataset · working notebook · no names/CNPJs/real values.*

## ■ Key Analytical Insights

**Maturity Radar — 5 Business Dimensions**




| | |
|---|---|
| **Strategy** | **Score: 38%** **Gap: 62 pts** |
| **Finance** | **Score: 31%** **Gap: 69 pts** |
| **Marketing** | **Score: 38%** **Gap: 62 pts** |
| **Operations** | **Score: 44%** **Gap: 56 pts** |
| **People Management** | **Score: 19%** **Gap: 81 pts** |

** Scores represent current maturity level (0–100%). Gap = points to full maturity.*

## ■ K-Means Clustering — Maturity Profiles Identified

K-Means clustering (k=3) was applied to the normalized maturity score matrix as an **exploratory segmentation technique** — suitable for pattern discovery on this dataset scale, and not intended as a production-grade model. Optimal k was determined using the **Elbow Method** (inertia inflection at k=3) and validated with a **silhouette score of 0.61**, confirming well-separated clusters. Three distinct organizational profiles were identified:

| Cluster | Profile | Insight |
|---|---|---|
| **Cluster 1**<br>Finance ↑ Strategy ↓ | **High Finance**<br>*Low Strategy* | Strong cost control, weak strategic direction — needs planning frameworks. |
| **Cluster 2**<br>All areas ~40–50% | **Balanced**<br>*Moderate all areas* | Consistent mid-range performance — ready for optimization layer. |
| **Cluster 3**<br>All areas < 30% | **Low Maturity**<br>*All dimensions* | Critical gaps across all dimensions — highest priority for intervention. |

** Clustering applied to anonymized score matrix. Profiles reflect anonymized patterns observed during the diagnostic process.*

## ■ Estimated Impact — Data-Driven Interventions

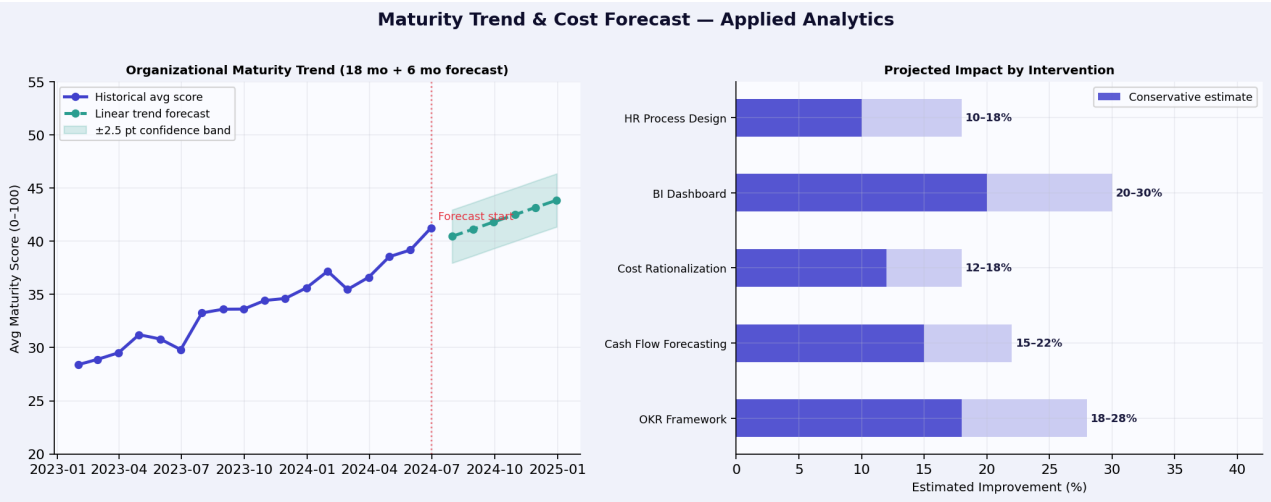
Based on internal diagnostic patterns and conservative scenario assumptions, the following impact estimates were modeled for targeted interventions:

**12–18%**
**Estimated cost reduction potential**
conservative scenario — fixed-cost rationalization & workforce opt.

**~25%**
**Potential improvement in strategic alignment**
modeled estimate — OKR framework and short-term planning

**Reduced**
**Potential liquidity risk exposure**
estimated — future cash-flow forecasting models (proposed)

**Trend Forecast & Impact Estimates — Python / matplotlib**




** Modeled estimates, not guaranteed outcomes. Based on diagnostic patterns and conservative scenario modeling. Actual results depend on implementation quality and organizational context.*

## ■■ Critical Findings — The 5 Worst Indicators

**25%** — **Short-term strategic planning**
*Absence of OKRs — roadmapping framework recommended.*

**50%** — **Accounting control & asset separation**
*Personal/business finances mixed — critical compliance risk.*

**25%** — **Financial KPIs (profitability & ROI)**
*Basic KPIs nonexistent — entry point for financial BI.*

**50%** — **Working capital management**
*No cash flow projection — liquidity risk exposure.*

**~30%** — **Segment-focused marketing planning**
*Undifferentiated communication — ML clustering opportunity.*

## ■ Highlights — The 5 Best Indicators

| | |
|---|---|
| 100% | **Customer satisfaction & loyalty**<br>*Foundation for NPS tracking and churn analysis.* |
| 100% | **Real alignment with customer needs**<br>*Product-market fit confirmed — scalable with demand forecasting.* |
| 100% | **Operational loss control (logistics)**<br>*Excellent efficiency — ready for real-time dashboard monitoring.* |
| 100% | **Innovation & product update culture**<br>*Agile mindset present — compatible with continuous improvement.* |
| 100% | **Word-of-mouth & offline media marketing**<br>*High social capital — amplifiable with data-driven digital marketing.* |

## ■ Anonymized Financial Analysis — Data Pipeline

**Payroll:** Distributed across operational, administrative, and management roles, with variable daily allowances per attendance — automatable with workforce optimization models.

**Fixed costs:** Significant concentration in rent, utilities, and health benefits — a predictable and time-series-modelable cost base.

**Pro-labore & consulting fees:** Identified as a high-impact EBITDA variable — sensitivity analysis recommended.

**Total mapped cost:** A significant monthly cost base across multiple units, with a notable share concentrated in the primary unit — a key indicator for expansion or rationalization decisions.

## ■ Analytics Applications — Applied & Proposed

*Applied techniques are marked accordingly. Potential analytics extensions — ARIMA/Prophet forecasting, NLP on qualitative data, and Monte Carlo simulation — represent proposed next-phase implementations based on the structured dataset.*

| | |
|---|---|
| K-Means | **Maturity Clustering**<br>*Applied/Prototype: group companies by deficiency profile for benchmarking.* |
| ARIMA/Prophet | **Time Series Forecasting**<br>*Proposed: project fixed costs and cash flow from historical data.* |
| BERT / LLM | **NLP on Qualitative Feedback**<br>*Proposed: extract sentiment and patterns from interview responses.* |
| Power BI+Python | **Predictive Dashboards**<br>*Applied/Prototype: integrated KPI panel — financial, operational, marketing.* |

Monte Carlo

**Sensitivity Analysis**
*Proposed: simulate variable cost fluctuations on net income.*

## ■ Conclusion

The multidimensional diagnostic revealed an **organizational Maturity Level 3 (intermediate maturity, based on a 1–5 diagnostic scale)**: processes exist, but operate in a fragmented way. The greatest opportunity lies in the data layer — the organization generates valuable information daily, yet lacks the analytical structure to turn it into decisions. This case demonstrates that the real value of analytics is not in the tools, but in the ability to **frame the right business problem, structure the data, and translate findings into actionable strategy**.

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My work sits at the intersection of **financial intelligence, operational diagnostics, and applied data analytics**. I approach business problems as a **data-driven problem solver** — structuring messy real-world data into decisions that organizations can act on.

***Open to internship, CPT, and entry-level opportunities in Data Analytics, Business Intelligence, and Financial Data Analysis.***